

## SeokHyeon (Lucas) Kong

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| CONTACT INFORMATION | Moore 102, University of Pennsylvania<br>200 S 33rd St, Philadelphia, PA 19104                                                                                                                                                                            | Website: <a href="https://lucasshkong.github.io">lucasshkong.github.io</a><br>Email: <a href="mailto:shkong@seas.upenn.edu">shkong@seas.upenn.edu</a> |
| RESEARCH INTERESTS  | My research interests lie in <i>Quantum Computer Architecture</i> , currently focusing on Quantum Error Correction.                                                                                                                                       |                                                                                                                                                       |
| EDUCATION           | <b>University of Pennsylvania</b> , Philadelphia, PA                                                                                                                                                                                                      | Aug 2026 – Present                                                                                                                                    |
|                     | <ul style="list-style-type: none"><li>• Ph.D. in Computer and Information Science</li><li>• Advisor: Prof. Gushu Li</li></ul>                                                                                                                             |                                                                                                                                                       |
|                     | <b>University of Pennsylvania</b> , Philadelphia, PA                                                                                                                                                                                                      | Aug 2026 – Present                                                                                                                                    |
|                     | <ul style="list-style-type: none"><li>• M.S. in Computer and Information Science</li></ul>                                                                                                                                                                |                                                                                                                                                       |
|                     | <b>Sungkyunkwan University (SKKU)</b> , Suwon, Korea                                                                                                                                                                                                      | Mar 2020 – Feb 2026                                                                                                                                   |
| PUBLICATIONS        | <ul style="list-style-type: none"><li>• B.S. in Electronic and Electrical Engineering</li><li>• B.S. in Advanced Semiconductor Engineering – System Architecture Track</li><li>• <i>Graduated with Summa Cum Laude</i> (GPA: 4.36/4.5)</li></ul>          |                                                                                                                                                       |
|                     | <b>The Pennsylvania State University</b> , University Park, PA                                                                                                                                                                                            | Fall 2024                                                                                                                                             |
|                     | <ul style="list-style-type: none"><li>• Exchange Student in Computer Engineering</li><li>• <i>Dean's List</i> (Grade: 4.0/4.0)</li></ul>                                                                                                                  |                                                                                                                                                       |
|                     | [1] Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy<br>SeokHyeon Kong, Donghwan Kim, EuiSeong Seo, and Kiwan Maeng<br><b>CAL 2025</b> – IEEE Computer Architecture Letters                              |                                                                                                                                                       |
|                     | [2] Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system<br>SeokHyeon Kong and Hayoung Oh<br><b>JKIICE 2023</b> – Journal of the Korean Institute of Information and Communication Engineering |                                                                                                                                                       |
| POSTER PRESENTATION | [3] Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices<br>SeokHyeon Kong, Chanhee Park, Jang-il Moon, and Denny Dahl<br><b>Qcenter</b> – Quantum Information Research Support Center, Korea                      |                                                                                                                                                       |
| EXPERIENCE          | <b>University of Pennsylvania</b> , Philadelphia, PA                                                                                                                                                                                                      | Aug 2026 – Present                                                                                                                                    |
|                     | <i>Graduate Research Fellow in the Penn Quantum System Lab</i>                                                                                                                                                                                            |                                                                                                                                                       |
|                     | <b>Hanwha Systems</b> , Seongnam, Korea                                                                                                                                                                                                                   | Jan 2026 – Jul 2026                                                                                                                                   |
|                     | <i>Software Engineer in the Avionics System R&amp;D Team</i> <ul style="list-style-type: none"><li>• Collaborated with The Boeing Company on ELAD (Eagle Large Area Display) for F-15 aircraft.</li></ul>                                                 |                                                                                                                                                       |

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|                           | <b>IonQ</b> , Remote<br><i>Research Mentee with Dr. Denny Dahl</i>                                                                                                                                                                                                                                                                                                             | Aug 2025 – Sep 2025                                                                              |
|                           | <ul style="list-style-type: none"> <li>Addressed the Instant Insanity problem by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods [3].</li> <li>Solved the Four Corners Map Coloring problem using a Quantum Approximate Optimization Algorithm (QAOA) with two different encoding schemes [3].</li> </ul>     |                                                                                                  |
|                           | <b>The Pennsylvania State University</b> , University Park, PA<br><i>Undergraduate Researcher with Prof. Kiwan Maeng</i>                                                                                                                                                                                                                                                       | Aug 2024 – Sep 2025                                                                              |
|                           | <ul style="list-style-type: none"> <li>Conducted GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differential privacy [1].</li> <li>Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality [1].</li> </ul> |                                                                                                  |
| TEACHING<br>EXPERIENCE    | <b>Academic Tutor</b> , SKKU<br><i>Intro to Electromagnetism</i>                                                                                                                                                                                                                                                                                                               | Fall 2023                                                                                        |
|                           | <ul style="list-style-type: none"> <li>Led weekly Q&amp;A sessions and problem-solving discussions.</li> </ul>                                                                                                                                                                                                                                                                 |                                                                                                  |
|                           | <b>Teaching Assistant</b> , SKKU<br><i>Engineering Mathematics 2</i>                                                                                                                                                                                                                                                                                                           | Summer 2023                                                                                      |
|                           | <ul style="list-style-type: none"> <li>Prepared and organized class materials, including exams.</li> </ul>                                                                                                                                                                                                                                                                     |                                                                                                  |
| HONORS AND<br>AWARDS      | <b>Graduate Fellowship and Research Stipend Award</b> , UPenn<br><b>Summa Cum Laude</b> , SKKU<br><b>Director's Award</b> – Quantum Hackathon, QCenter<br><b>President's List</b> , SKKU<br><b>Dean's List</b> , SKKU<br><b>Korea–U.S. Student Exchange Scholarship</b><br>in the field of high-tech industry (\$9,000), KIAT<br><b>Student Success Scholarship</b> , SKKU     | 2026<br>2026<br>2025<br>2024, 2025<br>SP2023, FA2023, SP2024, FA2024, SP2025<br>FA2024<br>FA2023 |
| SKILLS                    | <b>Software</b> : Qiskit, CUDA, C, Python, MATLAB, LaTeX<br><b>Hardware</b> : System Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute                                                                                                                                                                                                                                  |                                                                                                  |
| LEADERSHIP AND<br>SERVICE | <b>Executive Board</b> , President's List (Honor Society at SKKU)<br><b>Qiskit Advocate</b> , IBM Quantum<br><b>Squad Leader</b> , Republic of Korea Army (Military Service)                                                                                                                                                                                                   | 2025<br>2025<br>Apr 2021 – Oct 2022                                                              |
| REFERENCES                | <b>Gushu Li, Ph.D.</b><br>Assistant Professor of Computer and Information Science<br>University of Pennsylvania<br>Email: <a href="mailto:gushuli@seas.upenn.edu">gushuli@seas.upenn.edu</a>                                                                                                                                                                                   |                                                                                                  |
|                           | <b>Kiwan Maeng, Ph.D.</b><br>Assistant Professor of Computer Science and Engineering<br>The Pennsylvania State University<br>Email: <a href="mailto:kvm6242@psu.edu">kvm6242@psu.edu</a>                                                                                                                                                                                       |                                                                                                  |
|                           | <b>Denny Dahl, Ph.D.</b><br>Senior Director of Field Engineering<br>IonQ<br>Email: <a href="mailto:denny.dahl@ionq.co">denny.dahl@ionq.co</a>                                                                                                                                                                                                                                  |                                                                                                  |